

Owen Bress

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EDUCATION

The Pennsylvania State University, University Park, PA

GPA: 4.0/4.0 | Clubs: PSU Quants, Machine Learning PSU

Expected May 2028

Candidate for B.S. in Computer Science, B.S. in Mathematics, with minor in Statistics

Honors & Awards: The President Walker Award (for academic excellence), 6th Place at the TSA National Conference for Software Development 2023, Dean's List All Semesters, AP Scholar with Distinction Award

PROFESSIONAL EXPERIENCE

Strategic Websites | *Software Engineer Intern*, Bala Cynwyd, PA

June 2025 – August 2025

- Developed a TypeScript/Node.js REST API for automated visual regression testing across 100+ WordPress websites.
- Automated pre/post website updates using Puppeteer to flag optical differences of more than 0.5% for a rollback.
- Deployed on a VPS with Docker. Integrates GitHub Actions and Slack webhooks for a CI/CD pipeline + notifications.

Product Manager Accelerator | *Software Engineer Intern*, Remote

April 2025 – June 2025

- Collaborated with Project Managers and Designers to build and launch an AI personal assistant used by 500+ students.
- Supports automated onboarding, role-based FAQs, scheduled reminders of tasks, and an ATS-optimized resume review.
- Utilized React, Node.js, GPT API, the Discord Bot API, and developed in a trunk-based Git workflow.

Lower Merion Computer Science Club | *Officer*, Ardmore, PA

September 2021 – June 2024

- Planned, developed, and led an annual event to introduce STEM to students from underrepresented backgrounds.
- Traveled to teach middle & high school students Python and how to utilize it for flying drones, cars, path finding, etc.

PROJECTS

YouTube Dashboard

- Web app that manages multiple YouTube channels for users. Automatically ideates, generates, and schedules videos across channels. Between all users it brings in more than 2 million views per month and has monetized 10+ channels.
- Built the frontend using NextJS, Typescript, and Tailwind CSS. The backend server utilizes NodeJS, Express JS, Supabase Auth and Postgres. Uses AWS S3 buckets for media storage, and AWS lambda for video production.
- Technologies used include OpenRouter, ElevenLabs, Replicate, FFmpeg, YouTube Data, Vercel, Railway, and more.

CanvaAI

- Lead developer for a Chromium extension that utilizes a RAG system for a Canvas assistant compliant with FERPA. Processes 100+ documents using FastAPI, FAISS vector database, and sentence transformers for similarity search.
- Implements intelligent query routing with LangChain and Google Gemini API. Provides context-aware responses through conversation history and semantic/keyword search algorithms for optimized responses.
- Developed product roadmap & technologies, React chrome extension, and the Python backend FastAPI.

Marketplace Scraper

- CLI tool that scrapes Facebook Marketplace for profitable deals and sends alerts in real time. Allows filters for listings and estimates their price and demand using eBay and sends a report to Discord. Used to make over \$1,000 in profit.
- Developed in Python utilizing Selenium, BeautifulSoup, NumPy/Pandas, OpenRouter, and the Discord API.

Event LM

- An iOS app that organizes sporting and club events for students, parents, and faculty in a user-friendly manner.
- Made with Swift/Swift UI and used Firebase Realtime Database as the backend database.
- Won 6th Place at TSA National Conference for Software Development and Lower Merion Best App of 2023.

SKILLS

Programming Languages: Python, TypeScript/JavaScript, CSS, HTML, C/C++, Java, SQL

Frameworks & Libraries: NumPy, FastAPI, Pydantic, FFmpeg, Selenium, Playwright, Next.js, React, ExpressJS, TailwindCSS

Other: Git, SQL, AWS Cloud Computing, Supabase, MongoDB, Railway, Firebase Realtime Database, GPT Finetuning